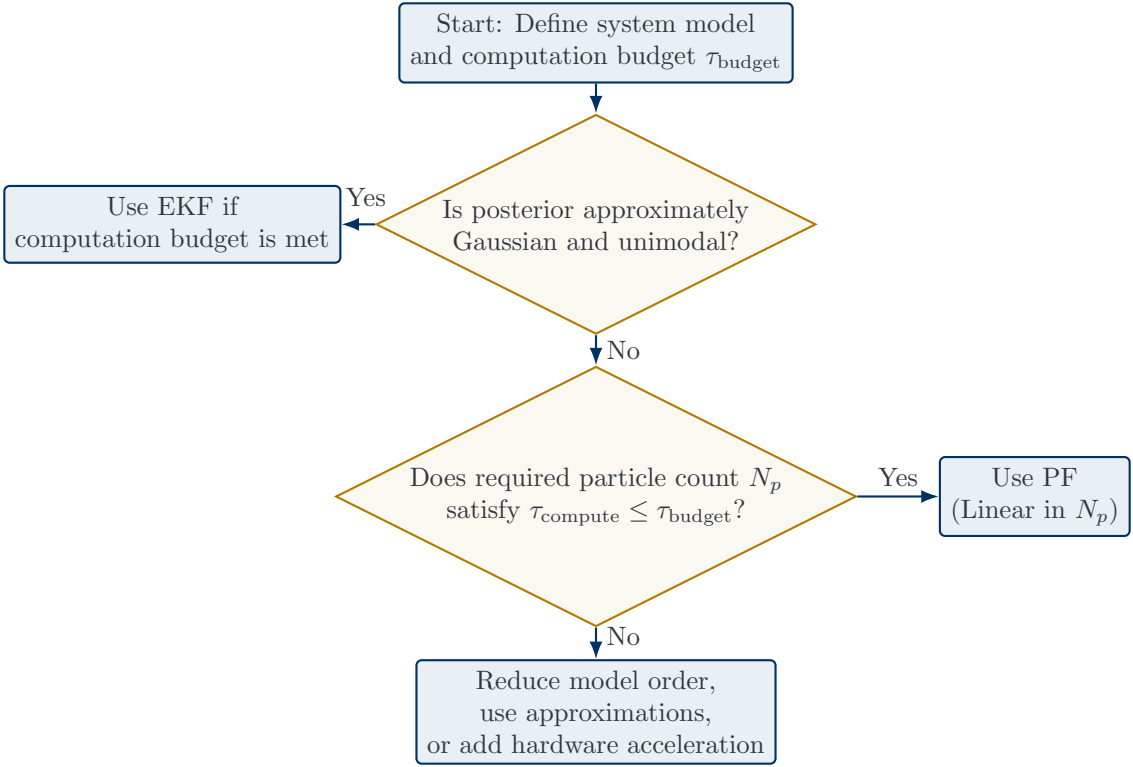
Start: Define system model and computation budget τ_{budget}
Is posterior approximately Gaussian and unimodal?
Yes
Use EKF if computation budget is met
No
Does required particle count N_p satisfy $\tau_{\text{compute}} \leq \tau_{\text{budget}}$?
Yes
Use PF (Linear in N_p)
No
Reduce model order, use approximations, or add hardware acceleration